

# Pavel Tokmakov

Senior Research Scientist, Toyota Research Institute

World models · Embodied AI · Robot learning · Video understanding

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pvtokmakov.github.io/home

West Hollywood, CA, USA

## RESEARCH EXPERIENCE

- 2020 – now **Senior Research Scientist** Los Altos, USA  
[Toyota Research Institute](#)  
Leading research on world models for embodied AI: learning predictive models of the physical world from large-scale video and leveraging them to advance robot learning.
- 2018 – 2020 **Postdoctoral Researcher** Pittsburgh, USA  
[CMU Robotics Institute](#) · with Prof. Martial Hebert and Prof. Deva Ramanan  
Working on problems related to video understanding and learning under limited supervision. Co-directing research projects, as well as conducting research individually.
- 2014 – 2018 **PhD Student** Grenoble, France  
[Inria THOTH team](#) · with Dr. Cordelia Schmid and Dr. Karteek Alahari  
Studying the role of motion in object recognition. The research resulted in both new methods for motion estimation and a novel approach for exploiting motion as a supervisory signal.

## EDUCATION

- 2014 – 2018 **PhD in Computer Science** France  
[University of Grenoble](#)  
Thesis: *Weakly-Supervised Learning for Object Recognition*
- 2012 – 2014 **MSc in Computer Science** Germany  
[University of Bonn](#)  
Emphasis on Intelligent Systems · GPA 4.0

## HONORS & AWARDS

- 2026 **PAMI Mark Everingham Prize**

## SELECTED PUBLICATIONS

\* Equal contribution

- Preprint**  
2026 **What 30,000 Hours of Ego-centric Video Does Not Teach**  
Jiahua Dong, Anurag Bagchi, Yash Jangir, Zubair Irshad, Sergey Zakharov, Martial Hebert, Homanga Bharadhwaj, Yu-Xiong Wang, Vitor Guizilini, **Pavel Tokmakov**  
*In submission*
- Preprint**  
2026 **GPWM: Generative Particle World Models for 3D Prediction and Planning**  
Robert Ren, Matthew Bronars, He Zhu, Sergey Zakharov, **Pavel Tokmakov**, Katerina Fragkiadaki  
*In submission*
- ECCV**  
2026 **Walk Through Paintings: Ego-centric World Models from Internet Priors**  
Anurag Bagchi, Zhipeng Bao, Homanga Bharadhwaj, Yu-Xiong Wang, **Pavel Tokmakov\***, Martial Hebert\*  
*European Conference on Computer Vision*
- ICLR**  
2026 **Capturing Visual Environment Structure Correlates with Control Performance**  
Jiahua Dong, Yunze Man, **Pavel Tokmakov\***, Yu-Xiong Wang\*  
*International Conference on Learning Representations*
- Sci. Rob.**  
2026 **A Careful Examination of Large Behavior Models for Multitask Dexterous Manipulation**  
TRI LBM Team  
*Science Robotics*

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| arXiv | <b>Video Generators Are Robust Robot Policies</b>                                                                                                                                                  |
| 2025  | Junbang Liang, <a href="#">Pavel Tokmakov</a> , Ruoshi Liu, Ege Ozguroglu, Sruthi Sudhakar, Paarth Shah, Rares Ambrus, Carl Vondrick<br><i>arXiv preprint</i>                                      |
| ICCV  | <b>ReferEverything: Towards Segmenting Everything We Can Speak of in Videos</b>                                                                                                                    |
| 2025  | Anurag Bagchi, Zhipeng Bao, Yu-Xiong Wang, <a href="#">Pavel Tokmakov</a> <sup>*</sup> , Martial Hebert <sup>*</sup><br><i>International Conference on Computer Vision</i>                         |
| ICML  | <b>Analyzing Question Complexity in Video QA via Visual Program Generation</b>                                                                                                                     |
| 2025  | Cristobal Eyzaguirre, Igor Vasiljevic, Achal Dave, Jiajun Wu, Thomas Kollar, Juan Carlos Niebles, <a href="#">Pavel Tokmakov</a><br><i>International Conference on Machine Learning</i>            |
| ECCV  | <b>Generative Camera Dolly: Extreme Monocular Dynamic Novel View Synthesis</b> ORAL                                                                                                                |
| 2024  | Basile Van Hoorick, Rundi Wu, Ege Ozguroglu, Kyle Sargent, Ruoshi Liu, <a href="#">Pavel Tokmakov</a> , Achal Dave, Changxi Zheng, Carl Vondrick<br><i>European Conference on Computer Vision</i>  |
| CoRL  | <b>Dreamitate: Real-world Visuomotor Policy Learning via Video Generation</b>                                                                                                                      |
| 2024  | Junbang Liang, Ruoshi Liu, Ege Ozguroglu, Sruthi Sudhakar, Achal Dave, <a href="#">Pavel Tokmakov</a> , Shuran Song, Carl Vondrick<br><i>Conference on Robot Learning</i>                          |
| CVPR  | <b>Understanding Video Transformers via Universal Concept Discovery</b> SPOTLIGHT                                                                                                                  |
| 2024  | Matthew Kowal, Achal Dave, Rares Ambrus, Adrien Gaidon, Konstantinos Derpanis, <a href="#">Pavel Tokmakov</a><br><i>Conference on Computer Vision and Pattern Recognition</i>                      |
| CVPR  | <b>pix2gestalt: Amodal Segmentation by Synthesizing Wholes</b> SPOTLIGHT                                                                                                                           |
| 2024  | Ege Ozguroglu, Ruoshi Liu, Didac Suris, Dian Chen, Achal Dave, <a href="#">Pavel Tokmakov</a> , Carl Vondrick<br><i>Conference on Computer Vision and Pattern Recognition</i>                      |
| ICCV  | <b>Zero-1-to-3: Zero-shot One Image to 3D Object</b>                                                                                                                                               |
| 2023  | Ruoshi Liu, Rundi Wu, Basile Van Hoorick, <a href="#">Pavel Tokmakov</a> , Sergey Zakharov, Carl Vondrick<br><i>International Conference on Computer Vision</i>                                    |
| CVPR  | <b>Breaking the “Object” in Video Object Segmentation</b>                                                                                                                                          |
| 2023  | <a href="#">Pavel Tokmakov</a> , Jie Li, Adrien Gaidon<br><i>Conference on Computer Vision and Pattern Recognition</i>                                                                             |
| ICML  | <b>Object Permanence Emerges in a Random Walk along Memory</b>                                                                                                                                     |
| 2022  | <a href="#">Pavel Tokmakov</a> , Allan Jabri, Jie Li, Adrien Gaidon<br><i>International Conference on Machine Learning</i>                                                                         |
| CVPR  | <b>Discovering Objects that Can Move</b>                                                                                                                                                           |
| 2022  | Zhipeng Bao <sup>*</sup> , <a href="#">Pavel Tokmakov</a> <sup>*</sup> , Allan Jabri, Yu-Xiong Wang, Adrien Gaidon, Martial Hebert<br><i>Conference on Computer Vision and Pattern Recognition</i> |
| ICCV  | <b>Learning to Track with Object Permanence</b>                                                                                                                                                    |
| 2021  | <a href="#">Pavel Tokmakov</a> , Jie Li, Wolfram Burgard, Adrien Gaidon<br><i>International Conference on Computer Vision</i>                                                                      |
| ECCV  | <b>TAO: A Large-Scale Benchmark for Tracking Any Object</b> SPOTLIGHT                                                                                                                              |
| 2020  | Achal Dave, Tarasha Khurana, <a href="#">Pavel Tokmakov</a> , Cordelia Schmid, Deva Ramanan<br><i>European Conference on Computer Vision</i>                                                       |
| ICCV  | <b>Learning Compositional Representations for Few-Shot Recognition</b>                                                                                                                             |
| 2019  | <a href="#">Pavel Tokmakov</a> , Yu-Xiong Wang, Martial Hebert<br><i>International Conference on Computer Vision</i>                                                                               |
| CVPR  | <b>A Structured Model For Action Detection</b>                                                                                                                                                     |
| 2019  | Yubo Zhang, <a href="#">Pavel Tokmakov</a> , Cordelia Schmid, Martial Hebert<br><i>Conference on Computer Vision and Pattern Recognition</i>                                                       |
| ICCV  | <b>Learning Video Object Segmentation with Visual Memory</b> ORAL                                                                                                                                  |
| 2017  | <a href="#">Pavel Tokmakov</a> , Karteek Alahari, Cordelia Schmid<br><i>International Conference on Computer Vision</i>                                                                            |
| CVPR  | <b>Learning Motion Patterns in Videos</b>                                                                                                                                                          |
| 2017  | <a href="#">Pavel Tokmakov</a> , Karteek Alahari, Cordelia Schmid<br><i>Conference on Computer Vision and Pattern Recognition</i>                                                                  |

## WORKSHOPS & ORGANIZATION

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Each workshop ran with an accompanying challenge. Responsibilities: proposals, speaker recruitment, program, panel moderation.

- 2024 – 2026 **VOTS Workshop Series** · ICCV / ECCV · co-organizer · video segmentation
- 2023 **5th LSVOS Workshop** · ICCV, Paris · lead organizer · video segmentation
- 2023 **2nd TAO Workshop** · CVPR, Vancouver · co-organizer · open-world tracking
- 2021 **RVSU Workshop** · CVPR · co-organizer · robust video scene understanding
- 2020 **TAO Workshop** · ECCV · lead organizer · tracking any object, new benchmark
- 2018 – 2020 **IARPA DIVA Project** · CMU · coordinated the CMU team on spatio-temporal action localization

## PROFESSIONAL SERVICE

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- 2015 – now **Area Chair / Reviewer** NeurIPS · ICLR · CVPR · ICCV · ECCV · ICML · CoRL
- 2019 – 2020 **CMU MSCV Admission Committee** Evaluating applications for the Master's in Computer Vision program

## LANGUAGES

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English (fluent) · Russian (native) · German (upper intermediate) · French (intermediate)